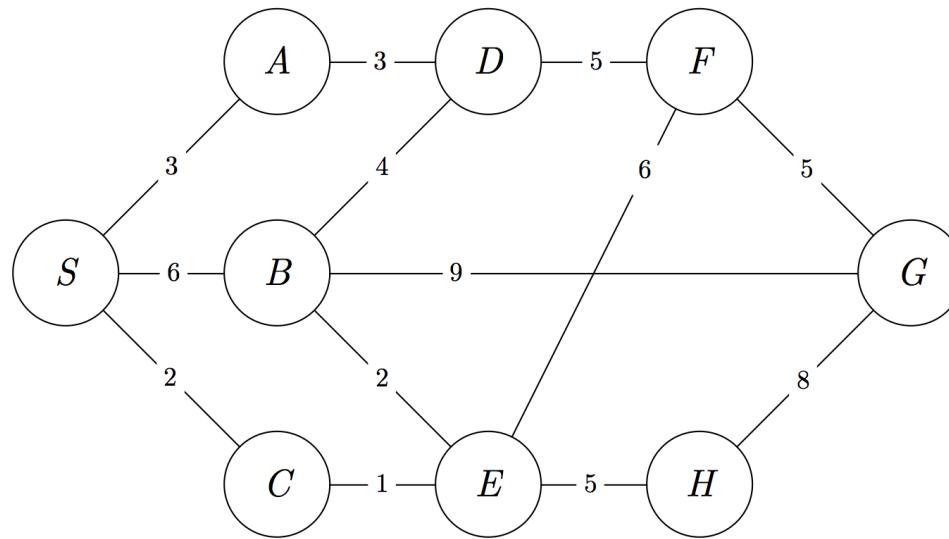
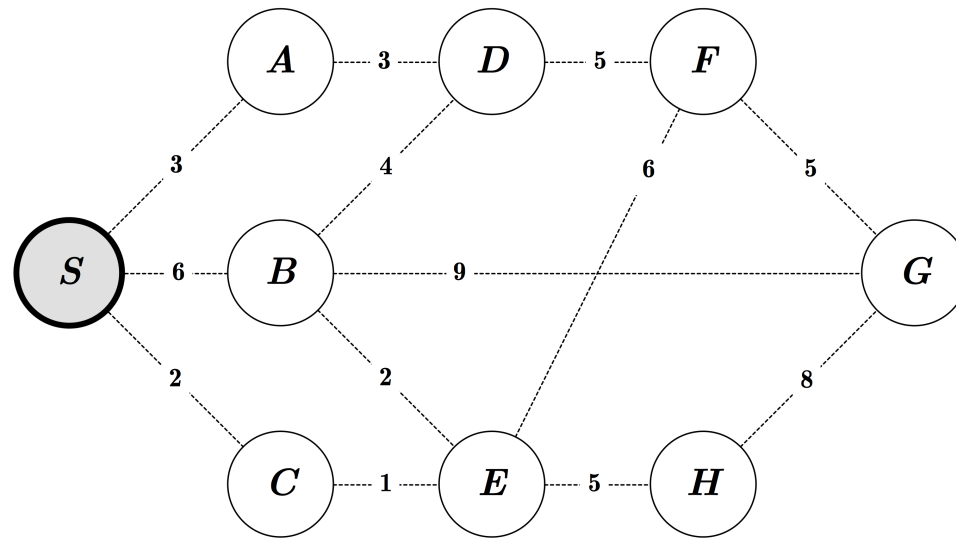


Exercise

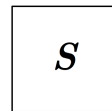


Question: What is the **order of visits of the nodes** and the **path** returned by BFS, DFS and UCS?

Exercise: BFS

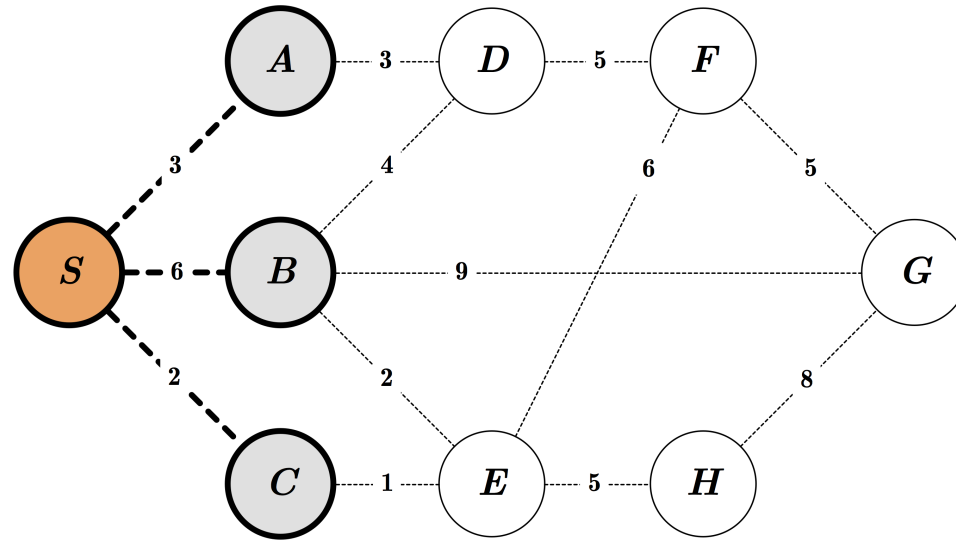


Queue:



Order of Visit:

Exercise: BFS



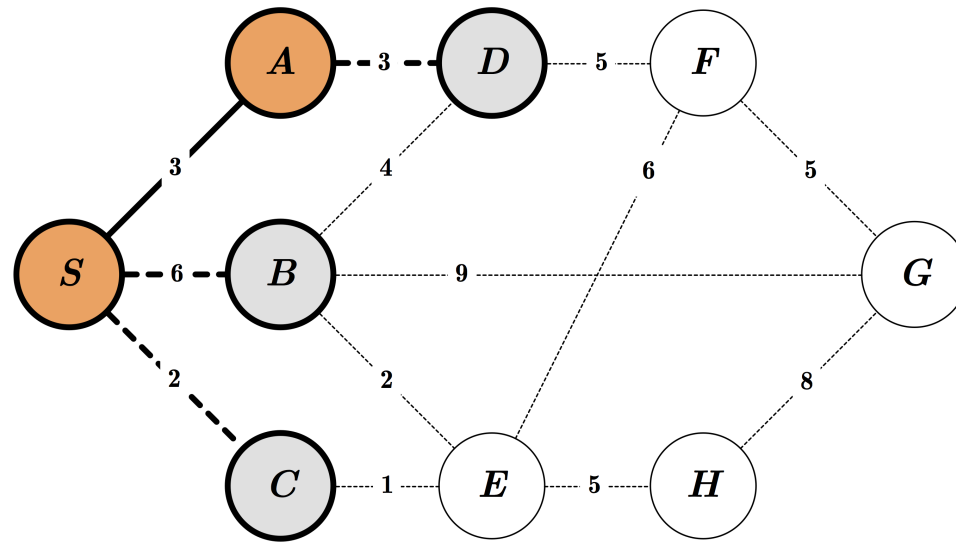
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>
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Order of Visit:

S

Exercise: BFS



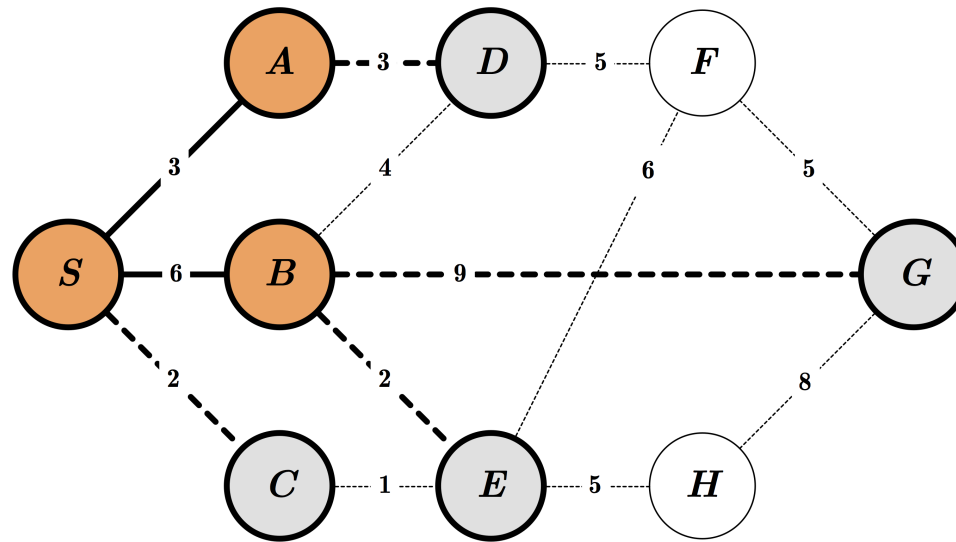
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
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Order of Visit:

S *A*

Exercise: BFS



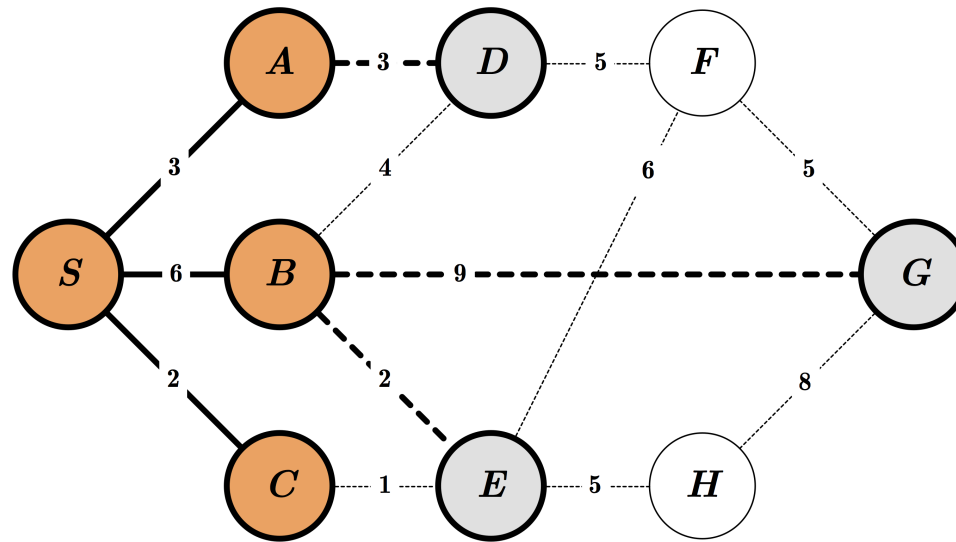
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>
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Order of Visit:

S *A* *B*

Exercise: BFS



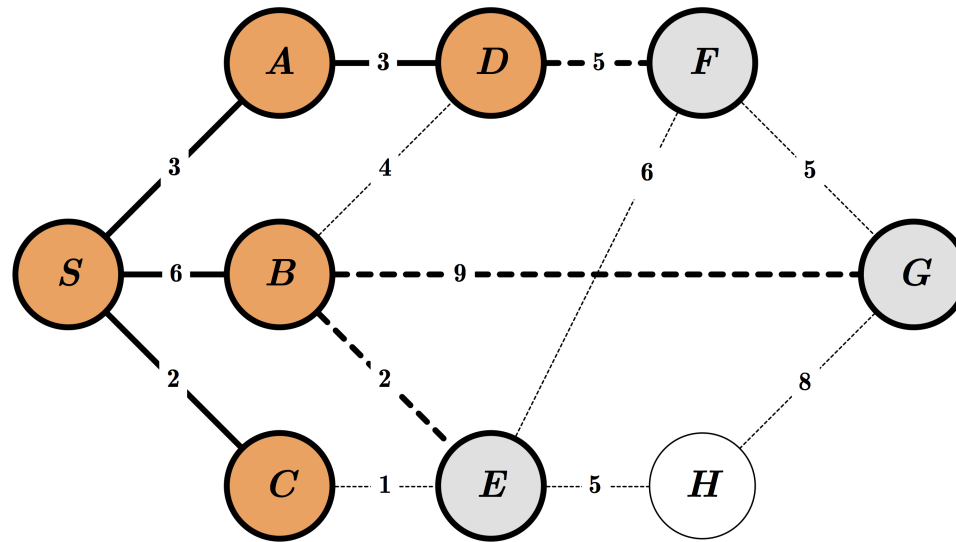
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>
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Order of Visit:

S *A* *B* *C*

Exercise: BFS



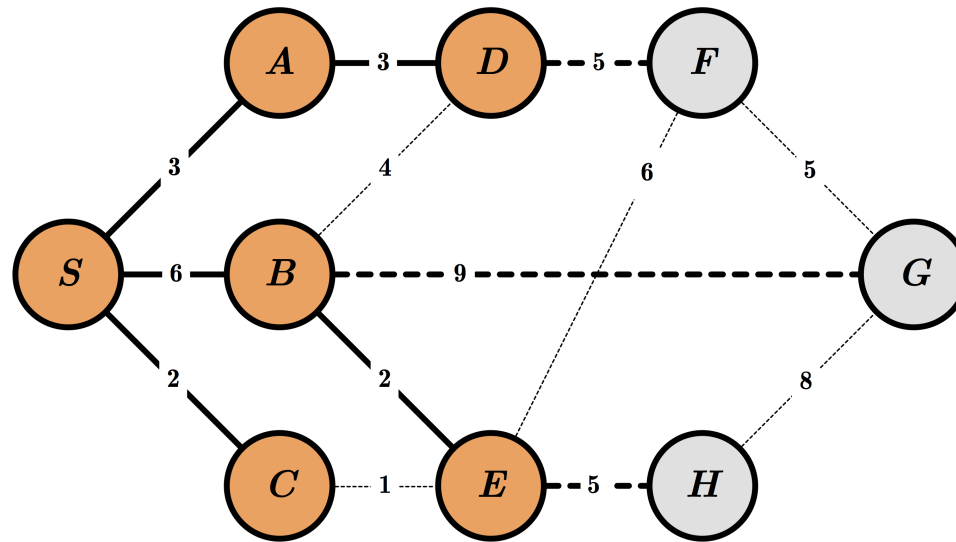
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>	<i>F</i>
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Order of Visit:

S *A* *B* *C* *D*

Exercise: BFS



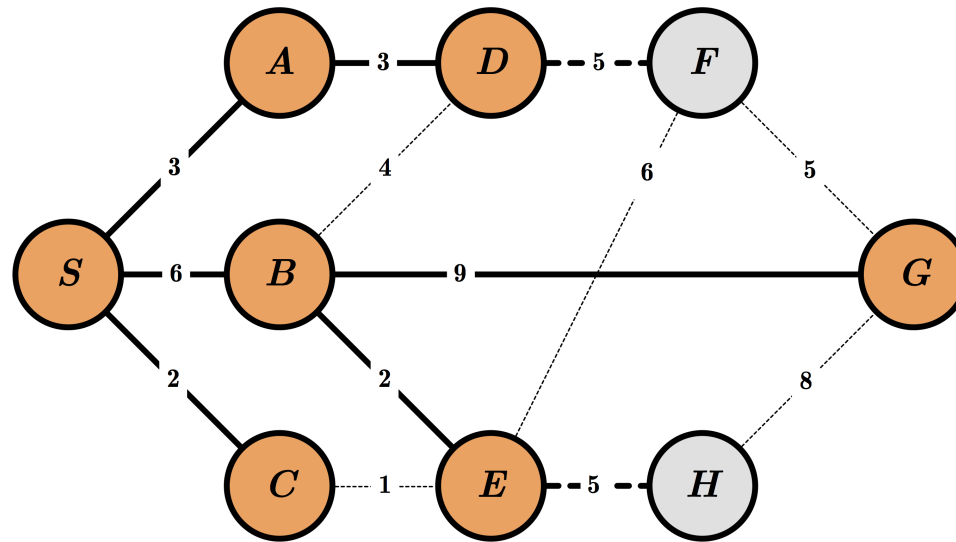
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>	<i>F</i>	<i>H</i>
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Order of Visit:

S *A* *B* *C* *D* *E*

Exercise: BFS



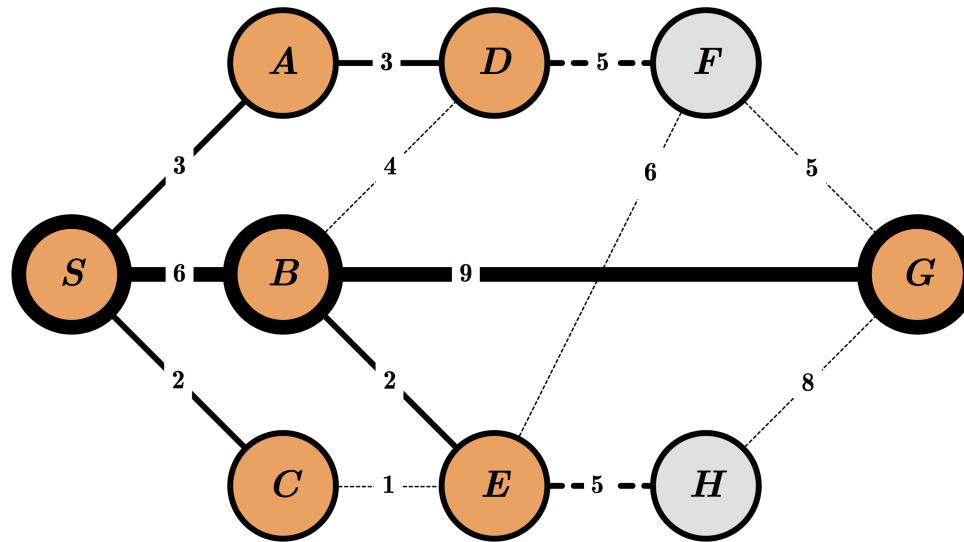
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>	<i>F</i>	<i>H</i>
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Order of Visit:

S *A* *B* *C* *D* *E* *G*

Exercise: BFS



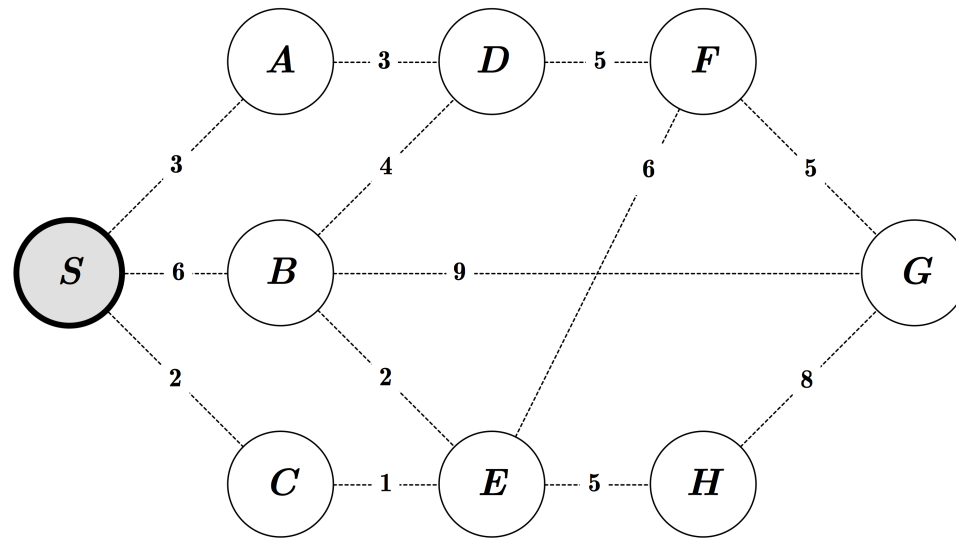
Queue:

<i>S</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>G</i>	<i>F</i>	<i>H</i>
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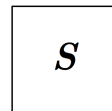
Order of Visit:

S *A* *B* *C* *D* *E* *G*

Exercise: DFS

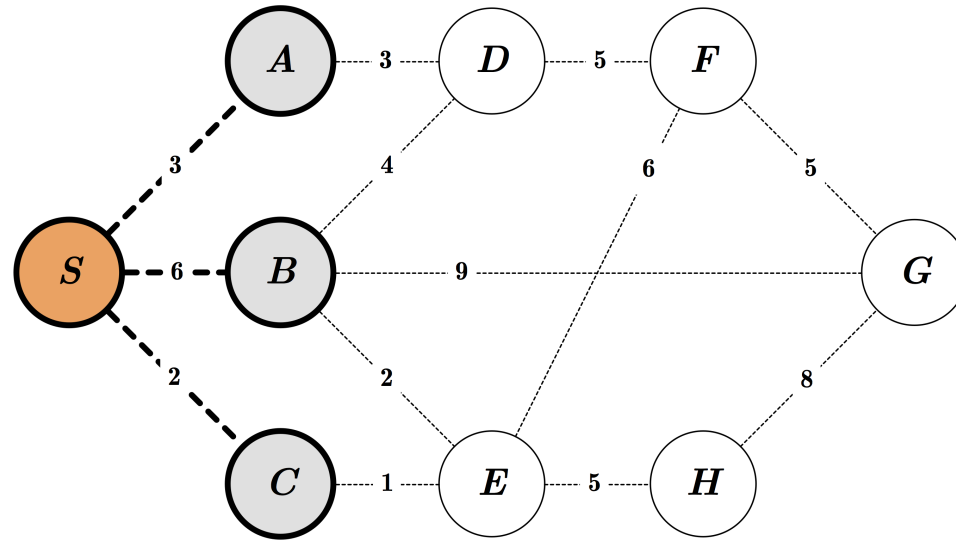


Stack:



Order of Visit:

Exercise: DFS



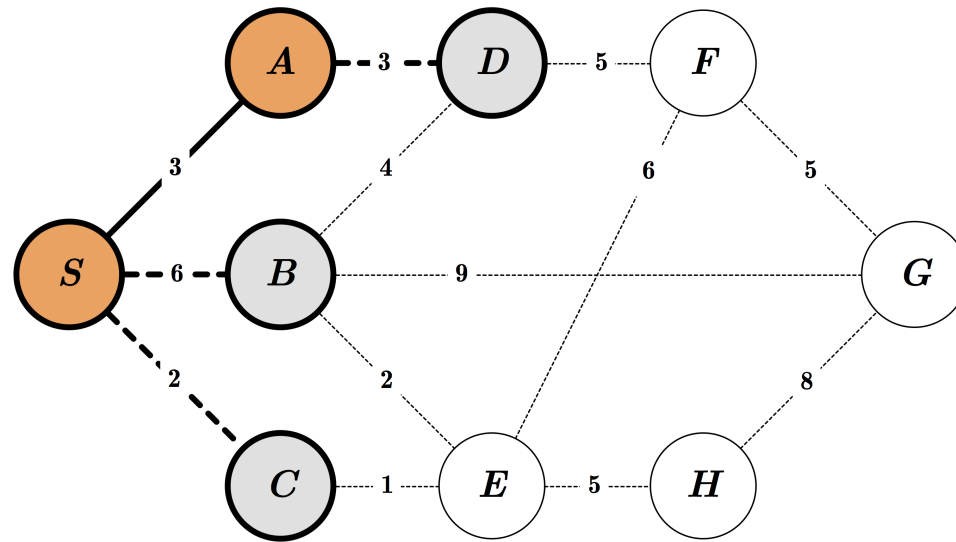
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>
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Order of Visit:

S

Exercise: DFS



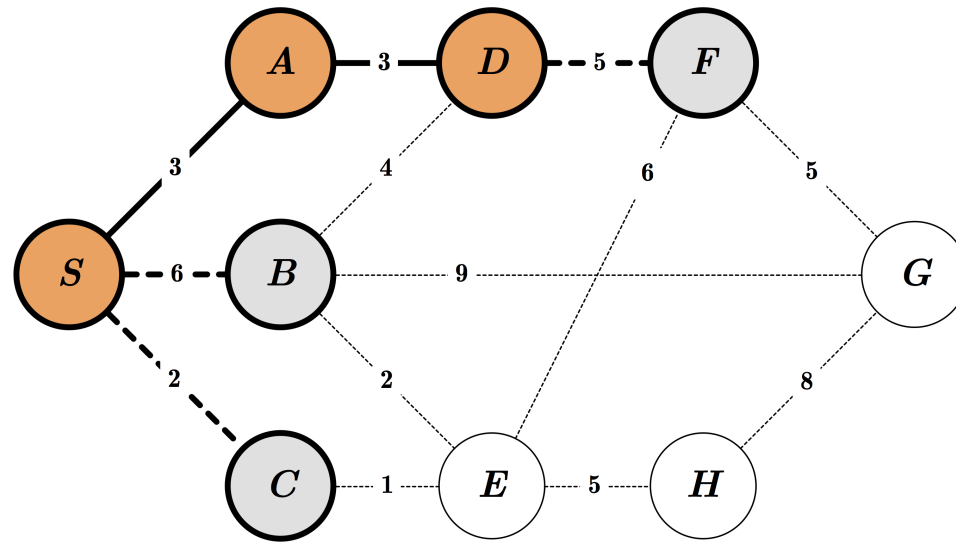
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>
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Order of Visit:

S *A*

Exercise: DFS



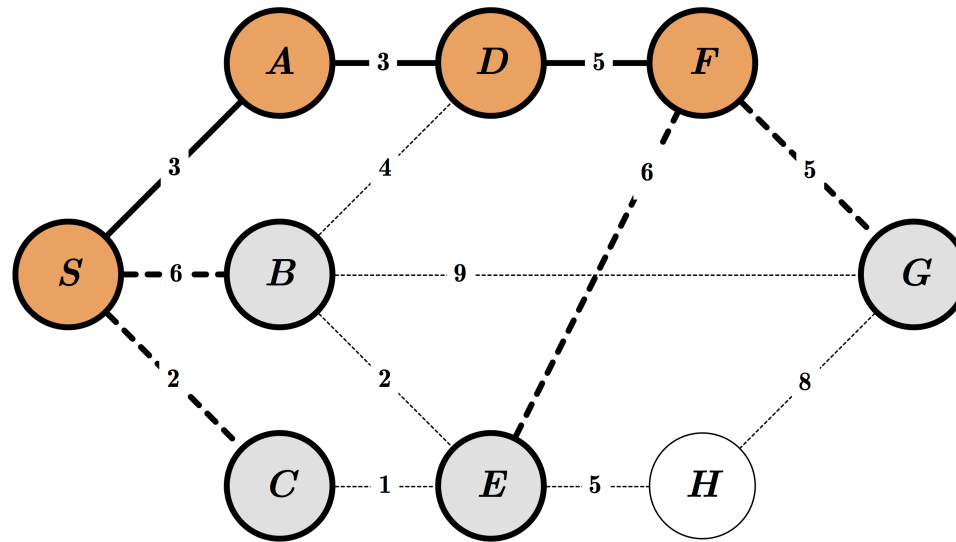
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>
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Order of Visit:

S *A* *D*

Exercise: DFS



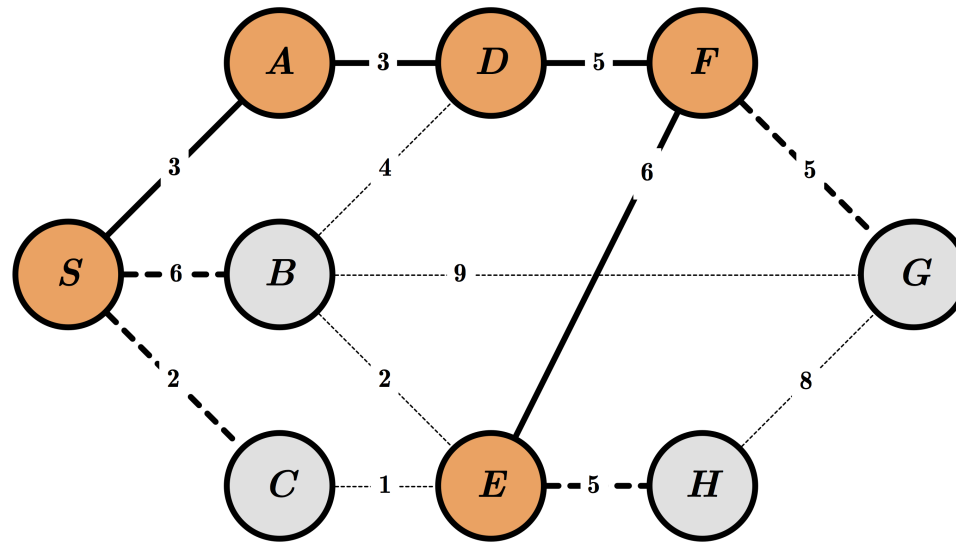
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>E</i>
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Order of Visit:

S *A* *D* *F*

Exercise: DFS



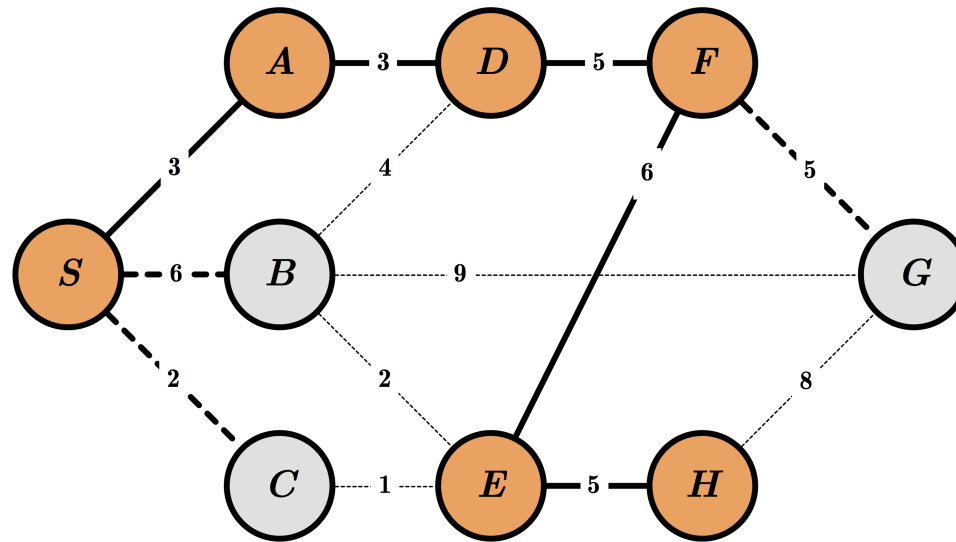
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>E</i>	<i>H</i>
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Order of Visit:

S *A* *D* *F* *E*

Exercise: DFS



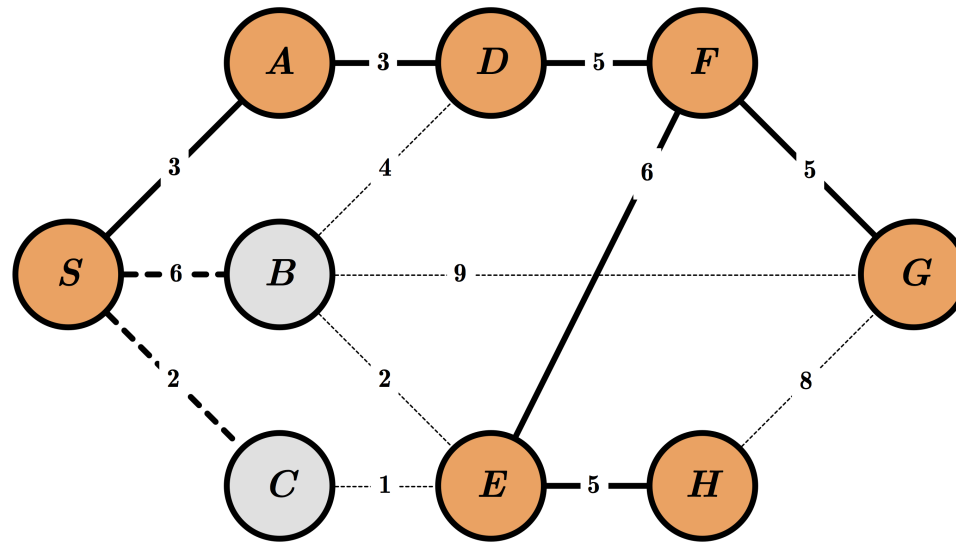
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>E</i>	<i>H</i>
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Order of Visit:

S *A* *D* *F* *E* *H*

Exercise: DFS



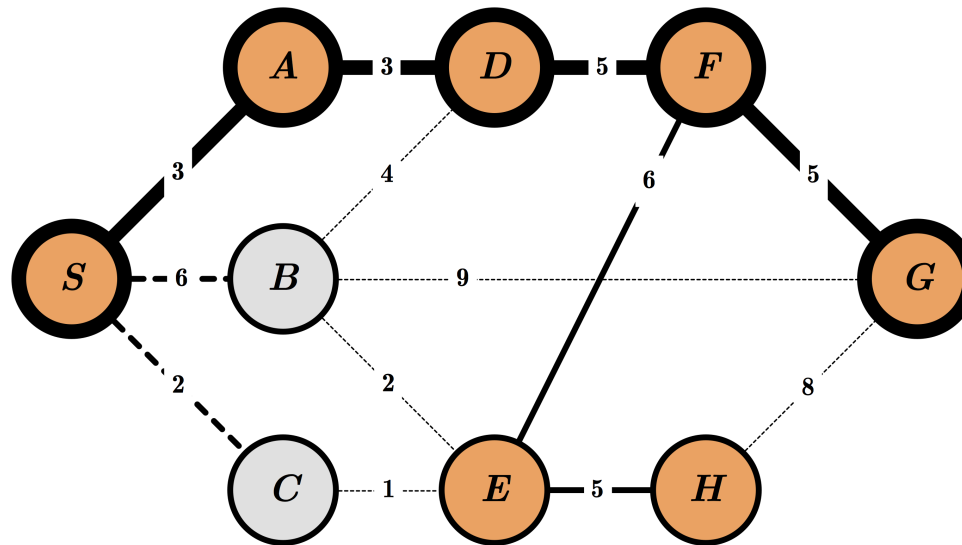
Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>E</i>	<i>H</i>
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Order of Visit:

S *A* *D* *F* *E* *H* *G*

Exercise: DFS



Stack:

<i>S</i>	<i>C</i>	<i>B</i>	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>E</i>	<i>H</i>
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Order of Visit:

S *A* *D* *F* *E* *H* *G*